

11

PUBLICATIONS

163

CITATIONS

6

H-INDEX

5

I10-INDEX

RESEARCH SUMMARY

PhD graduate (PPKE ITK, 2023) with six years of research experience at the Machine Perception Laboratory, HUN-REN SZTAKI. My doctoral dissertation, "Deep Learning Based Methods on 2D Image and 3D Structure Inpainting", applies AI techniques to 3D point cloud completion, geometry processing, image & scene analysis, and multispectral remote sensing. Actively seeking **research scientist** and **AI/ML engineer** positions to apply deep learning and 3D vision expertise to real-world problems.

EDUCATION

PhD in Information Technology & Bionics

2017 – 2023

Pázmány Péter Catholic University (PPKE ITK), Budapest, Hungary

Dissertation: "Deep Learning Based Methods on 2D Image and 3D Structure Inpainting" · Supervised by Prof. Csaba Benedek, Machine Perception Laboratory, HUN-REN SZTAKI

Master's in Computer Engineering

2013 – 2016

Tishreen University, Latakia, Syria

Thesis: "Design of 3D Virtual Environment Building System by RGB-D Camera on Vehicle"

Bachelor's in Computer Engineering

2009 – 2013

Tishreen University, Latakia, Syria

Thesis: "Tracking a Moving Object Using Laser with Image Processing Algorithms"

RESEARCH INTERESTS

- 3D Scene Reconstruction: Point Clouds, NeRF, 3D Gaussian Splatting
 - GAN-Based Image & Scene Inpainting
 - Deep Learning: CNN, GAN, Transformer, Autoregressive Models
- Remote Sensing & Geospatial AI (Sentinel-1/2)
 - Architectural & Heritage Image Analysis
 - Brain-Computer Interfaces (EEG Classification)

SELECTED PUBLICATIONS

Inland Excess Water (IEW) Monitoring Using Sentinel-1/2: A SplitClass Segmentation and Temporal Gap-Filling Approach

Y. Ibrahim, M. Belényesi, C. Liu, M. Richter-Cserey, M. Simon, T. Sziranyi, C. Benedek

ICCVW — IEEE/CVF, SEA Workshop · 2025

MVPCC-Net: Multi-View Based Point Cloud Completion Network for MLS Data

Y. Ibrahim, C. Benedek

Image and Vision Computing — Elsevier · 2023 · 11 citations

Deep Learning-Based Masonry Wall Image Analysis

Y. Ibrahim, B. Nagy, C. Benedek

Remote Sensing — MDPI · 2020 · 39 citations

A GAN-Based Blind Inpainting Method for Masonry Wall Images

Y. Ibrahim, B. Nagy, C. Benedek

IEEE Winter Conference on Applications of Computer Vision (WACV) · 2021

Multi-View Based 3D Point Cloud Completion Algorithm for Vehicles in Autonomous Driving

Y. Ibrahim, B. Nagy, C. Benedek

ICPR — IEEE · 2022 · 7 citations

CNN-Based Watershed Marker Extraction for Brick Segmentation in Masonry Walls

Y. Ibrahim, B. Nagy, C. Benedek

International Symposium on Visual Computing — Springer · 2019

Multi-Class Classification of Motor Imagery EEG Signals Using Image-Based Deep Recurrent Convolutional Neural Network

W. Fadel, C. Kollod, M. Wahdow, Y. Ibrahim, I. Ulbert

8th International BCI Meeting — IEEE · 2020 · 57 citations

Full list: scholar.google.com/citations?user=1ShbTvIAAAAJ

TECHNICAL SKILLS

Languages: Python, MATLAB, C++

ML / DL Frameworks: PyTorch, TensorFlow, Keras, PyTorch Lightning

Architectures: Transformer, Autoregressive, CNN, GAN

Computer Vision & 3D: OpenCV, Open3D, PCL, scikit-image, NeRF, 3D Gaussian Splatting

Research & Tools: ROS2, Weights & Biases, Jupyter, LaTeX, Git, Docker, Linux/HPC

PROFESSIONAL SERVICE

Peer reviewer at 4 Elsevier journals · 9 reviews · [Verified on ORCID](#)

Engineering Applications of AI (Elsevier Q1, 2025)

Automation in Construction (Elsevier Q1, 2024)

Measurement (Elsevier Q1, 2024)

NOTABLE PROJECTS

3D Point Cloud Completion for Robotic Perception

github.com/sztaki-geocomp/Multi-view-3d-completion

Multi-view deep learning approach for completing partial LiDAR point clouds caused by occlusion in autonomous driving and robotic perception.

Satellite Intelligence: Inland Flood & Water Monitoring

waterdetection.github.io · [HuggingFace Demo](#)

SplitClass segmentation network for detecting and tracking inland excess water using Sentinel-1/2 satellite imagery (ICCVW 2025).

AI-Powered Masonry Wall Analysis & Restoration

mplab.sztaki.hu/geocomp/masonryWallAnalysis

GAN-based blind inpainting model for segmenting bricks, detecting damaged regions, and inpainting heritage masonry walls.

VOLUNTEER WORK

Translator — Science & Technology

Nov 2016 – Present

[Syrian Researchers](#)

Translating scientific and technical content to support knowledge dissemination within the Syrian research community.